

(a)

(b)

(c)

(d)

UAE · Rx1day

UAE · R95p

Eastern · Rx1day

Eastern · R95p

Process Fidelity Score **A**

+0.14 +0.35 +0.09 +0.00

+0.11 +0.47 -0.01 -0.04

+0.12 +0.34 +0.06 -0.02

+0.04 +0.39 -0.11 -0.18

Rx1day climatology

-0.76 **-0.29** **-0.69** -0.32**-0.73** -0.16 -0.61 -0.26**-0.78** -0.32 **-0.72** -0.30**-0.77** -0.24 -0.65 -0.22

R95p climatology

-0.71 -0.10 -0.59 -0.08**-0.72** -0.11 -0.62 -0.10**-0.79** -0.07 -0.58 +0.01**-0.81** -0.06 -0.60 -0.02

R99p climatology

-0.67 -0.02 -0.55 +0.01

-0.68 -0.05 -0.58 -0.03

-0.69 -0.01 -0.51 +0.07

-0.70 +0.01 -0.50 +0.07

B
Tail concentration R95p/PRCPTOT

-0.58 +0.12 -0.47 +0.13

-0.61 +0.10 -0.54 +0.14

-0.79 -0.02 -0.58 +0.00**-0.79** +0.04 -0.57 +0.06

Event duration Rx5day/Rx1day

+0.26 -0.23 +0.19 -0.08

+0.43 -0.32 +0.30 -0.12

+0.46 -0.07 +0.36 +0.06

+0.56 -0.19 +0.41 -0.02

Wet-day frequency

-0.18 -0.17 -0.07 -0.05

-0.17 -0.27 -0.12 -0.17

-0.17 +0.03 +0.10 +0.17

-0.14 -0.01 -0.00 +0.02

Interannual variability, Rx1day

-0.50 +0.18 -0.42 +0.17

-0.52 +0.12 -0.48 +0.18

-0.70 +0.01 -0.55 +0.03

-0.71 +0.04 -0.54 +0.09

Interannual variability, R95p

-0.36 +0.32 -0.21 +0.34

-0.40 +0.16 -0.33 +0.24

-0.42 +0.43 -0.20 +0.43

-0.51 +0.32 -0.30 +0.34

C
Historical trend, Rx1day

-0.12 -0.18 -0.01 -0.08

-0.10 -0.16 +0.07 -0.02

+0.01 +0.08 +0.14 -0.03

+0.00 +0.15 +0.24 +0.06

Historical trend, R95p

-0.12 -0.13 -0.01 -0.12

-0.12 -0.10 -0.00 -0.06

-0.08 +0.02 -0.09 -0.25

-0.08 +0.01 -0.03 -0.20

SSP1-2.6
SSP2-4.5
SSP3-7.0
SSP5-8.5SSP1-2.6
SSP2-4.5
SSP3-7.0
SSP5-8.5SSP1-2.6
SSP2-4.5
SSP3-7.0
SSP5-8.5SSP1-2.6
SSP2-4.5
SSP3-7.0
SSP5-8.5

Correlation with projected absolute change



Bold outline and asterisk: satisfies all pre-specified criteria ($q < 0.05$, leave-one-family-out skill > 0 , sign stability > 0.9). Families: A process fidelity, B present-day extreme state, C variability and historical change.